AWS re:Invent 2025 re:Cap

Subin Kim

발표자

김수빈

- AWS Community Builder (2023~)
- AWSKRUG 플랫폼 엔지니어링 소모임 오거나이저
- 당근마켓에서 공통 서비스 플랫폼 개발을 하고 있습니다.



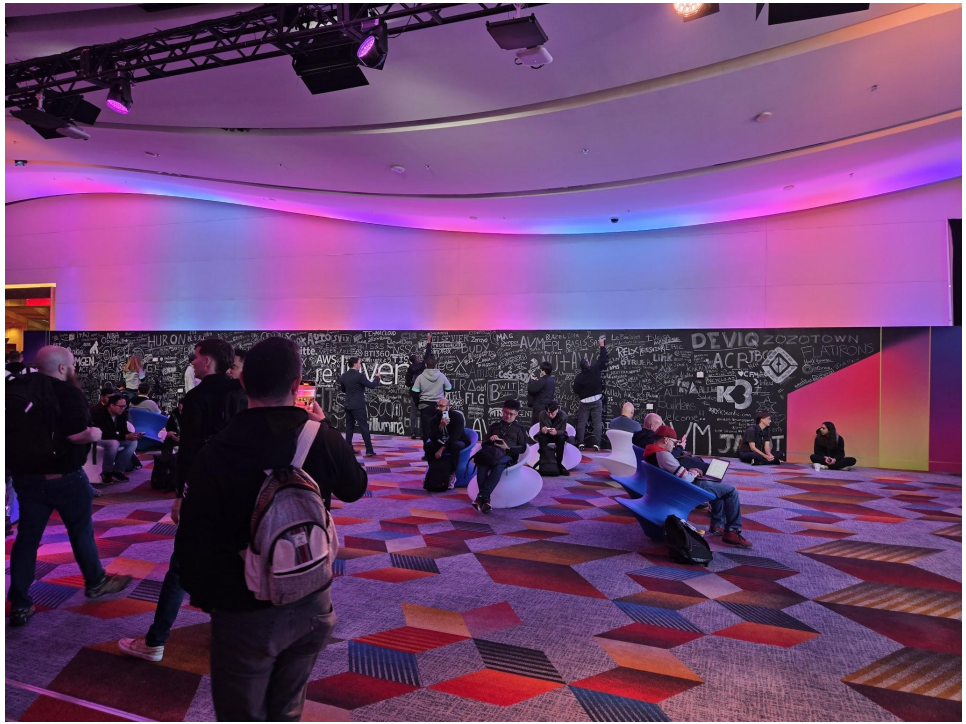
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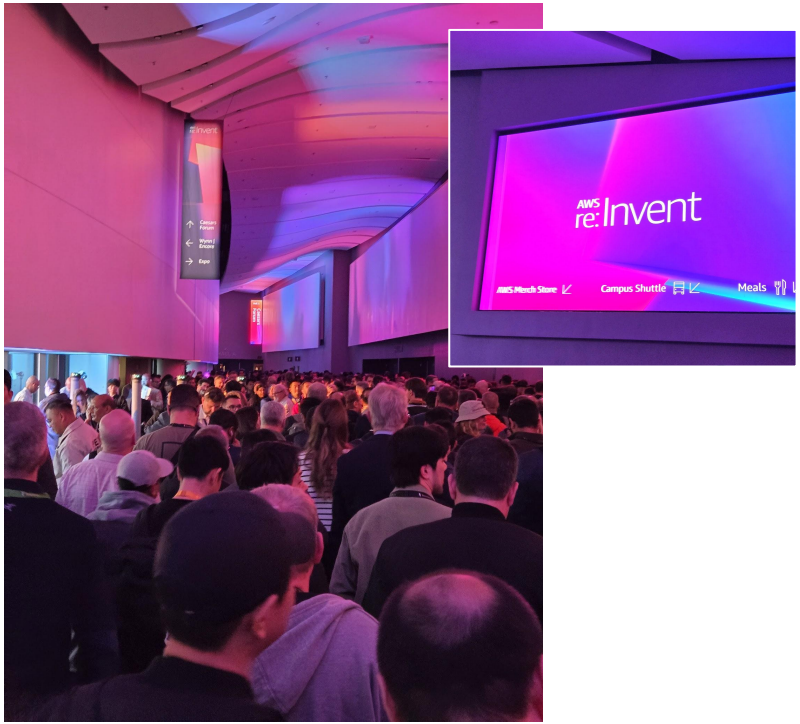
AWS re:Invent란?

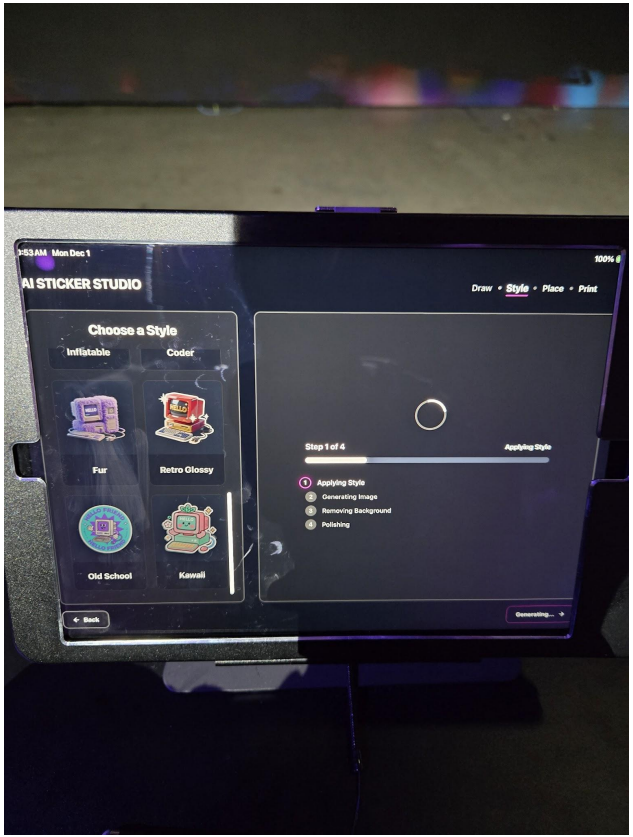
- AWS re:Invent는 AWS가 매년 개최하는 세계 최대 규모의 클라우드 컨퍼런스
- AWS의 최신 기술 발표, 키노트 강연, 수천 개의 발표 및 핸즈온 세션과 엑스포, 네트워킹 기회가 제공됩니다.

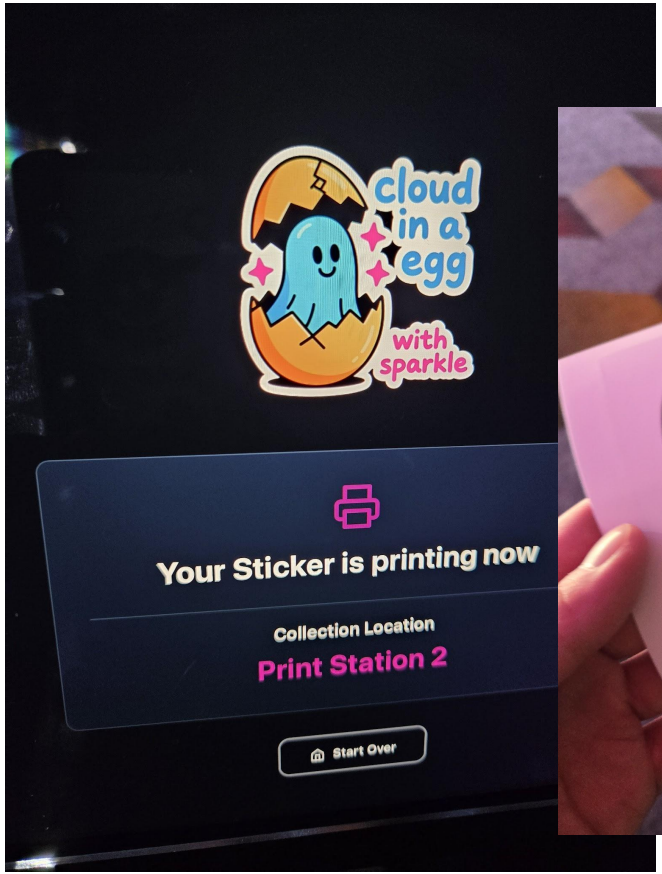


AWS re:Invent란?

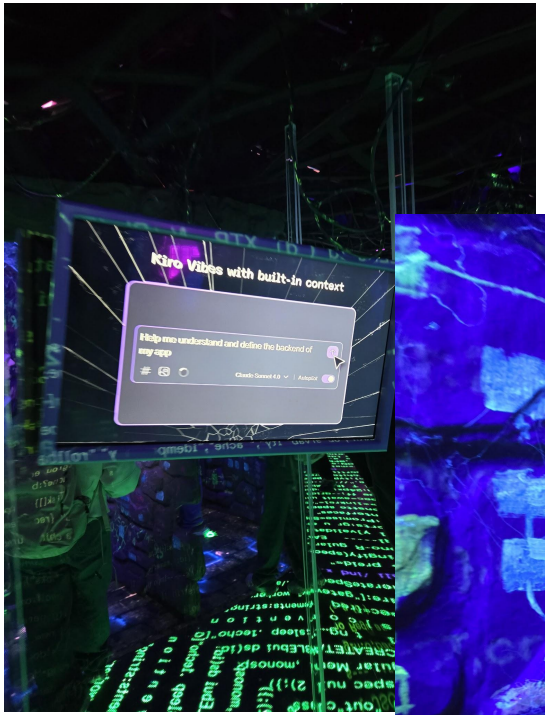




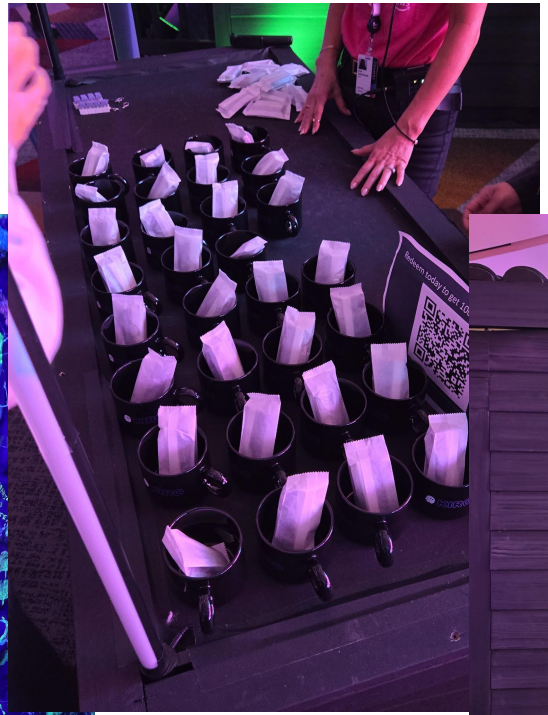

AI STICKER STUDIO
Draw • Style • Place • Print
Choose a Style
Inflatable
Coder
Fur
Retro Glossy
Old School
Kawaii
Step 1 of 4
Applying Style
Applying Style
Generating Image
Removing Background
Polishing
Back


cloud in a egg with sparkle
Your Sticker is printing now
Collection Location
Print Station 2
Start Over


cloud in a egg with sparkle


Kiro Vibes with built-in context
Help me understand and define the backend of my app






I SURVIVED
THE HOUSE OF
KIRO

Build production AI agents with the Strands Agents SDK for TypeScript

총 12개의 세션

AIM3309 ☆ 완료

Slack securely powers internal AI dev tools with Bedrock and Strands

EN KO JA

AIM426 ☆ 완료

Using Strands Agents to build autonomous, self-improving AI agents

EN KO JA

AIM3331 ☆ 완료

Build production AI agents with the Strands Agents SDK for TypeScript

EN KO JA

DEV330 ☆ 완료

Build a multi-agent role-playing Game Master with Strands Agents

EN KO JA

NTA406 ☆ 완료

Observability for Reliable Agentic AI with Strands SDK & OpenTelemetry

EN KO JA

NTA405 ☆ 완료

Building AI Agents with Serverless, Strands, and MCP

EN KO JA

DEV332 ☆ 완료

Build multi-modal AI agents with Strands Agents and Amazon S3 Vectors

EN KO JA

NTA4051 ☆ 완료

Building AI Agents with Serverless, Strands, and MCP

EN KO JA

DEV327 ☆ 완료

Building Agentic AI: Amazon Nova Act and Strands Agents in Practice

EN KO JA

DEV326 ☆ 완료

Anti-Money Laundering Multi-agent Orchestration with AWS Strands

EN KO JA

SPF304 ☆ 완료

NFL Fantasy AI: Zero to Production in Weeks with Bedrock and Strands Agents

EN KO JA

AIM233 ☆ 완료

Build observable AI agents with Strands, AgentCore, and Datadog

EN KO JA

AWS의 AI 서비스 역사



Amazon SageMaker



Amazon Personalize



Amazon Forecast



Amazon Nova



Amazon Bedrock

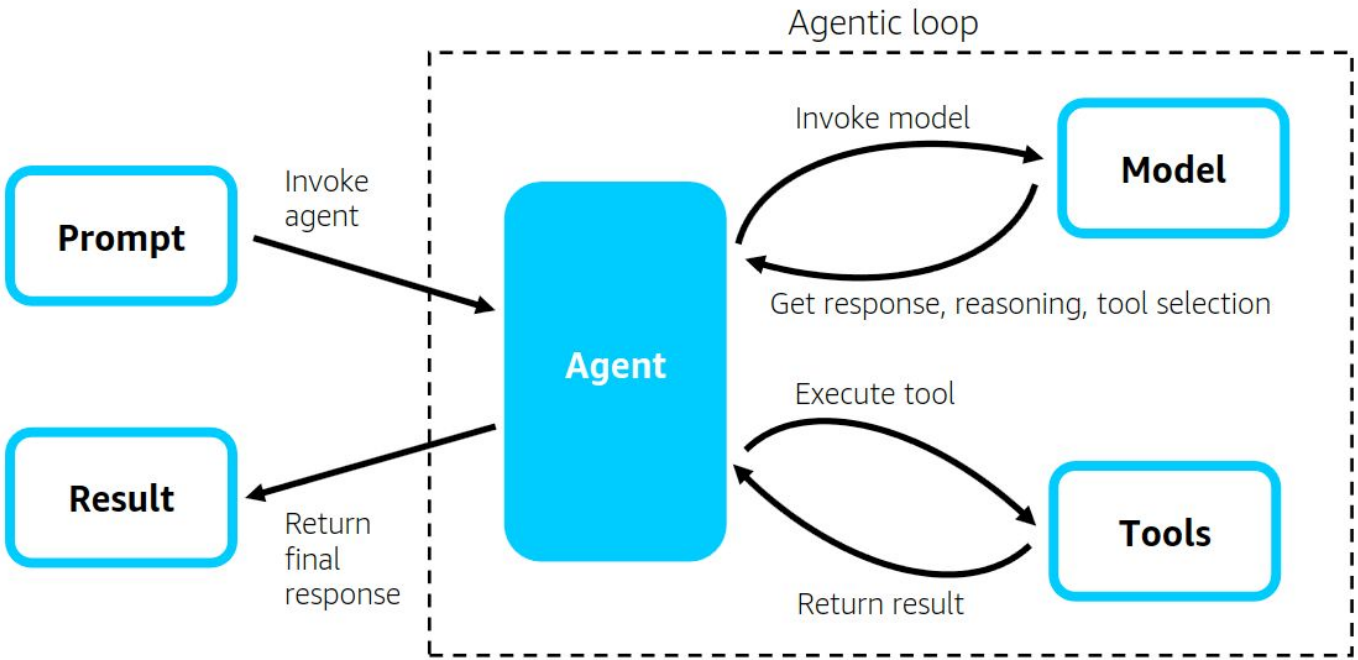


Amazon AgentCore



2017 2019 2023 2024 2025

Strands


Agentic loop
Prompt
Invoke agent
Agent
Invoke model
Model
Get response, reasoning, tool selection
Execute tool
Tools
Return result
Result
Return final response

AgentCore와 Strands

- AgentCore(인프라) 위에 올라가는 Strands(프레임워크)
- AgentCore 위에는 다른 프레임워크도 배포 가능
- Strands는 다른 인프라에도 배포 가능

Announcing TypeScript support in Strands Agents (preview) and more

Posted on: Dec 3, 2025

In May, we open sourced the Strands Agents SDK, an open source python framework that takes a model-driven approach to building and running AI agents in just a few lines of code. Today, we're announcing that TypeScript support is available in preview. Now, developers can choose between Python and TypeScript for building Strands Agents.

TypeScript support in Strands has been designed to provide an idiomatic TypeScript experience with full type safety, async/await support, and modern JavaScript/TypeScript patterns. Strands can be easily run in client applications, in browsers, and server-side applications in runtimes like AWS Lambda and Bedrock AgentCore. Developers can also build their entire stack in Typescript using the AWS CDK.

We're also announcing three additional updates for the Strands SDK. First, edge device support for Strands Agents is generally available, extending the SDK with bidirectional streaming and additional local model providers like llama.cpp that let you run agents on small-scale devices using local models. Second, Strands steering is now available as an experimental feature, giving developers a modular prompting mechanism that provides feedback to the agent at the right moment in its lifecycle, steering agents toward a desired outcome without rigid workflows. Finally, Strands evaluations is available in preview. Evaluations gives developers the ability to systematically validate agent behavior, measure improvements, and deploy with confidence during development cycles.

Head to the Strands Agents GitHub to get started building.

User Guide

Quickstart

Getting Started

Python

TypeScript

Concepts

Agents

Tools

Model Providers

Streaming

Multi-agent

Interrupts

Experimental

Safety & Security

Responsible AI

Guardrails

Prompt Engineering

PII Redaction

TypeScript Quickstart

> **Experimental SDK**
>
> The TypeScript SDK is currently experimental. It does not yet support all features available in the Python SDK, and breaking changes are expected as development continues. Use with caution in production environments.

This quickstart guide shows you how to create your first basic Strands agent with TypeScript, add built-in and custom tools to your agent, use different model providers, emit debug logs, and run the agent locally.

After completing this guide you can integrate your agent with a web server or browser, evaluate and improve your agent, along with deploying to production and running at scale.

Install the SDK

First, ensure that you have Node.js 20+ and npm installed. See the npm documentation for installation instructions.

On this page

Install the SDK

Configuring Credentials

Project Setup

Running Agents

Understanding What Agents Did

Console Output

Model Providers

Identifying a configured model

Amazon Bedrock (Default)

Additional Model Providers

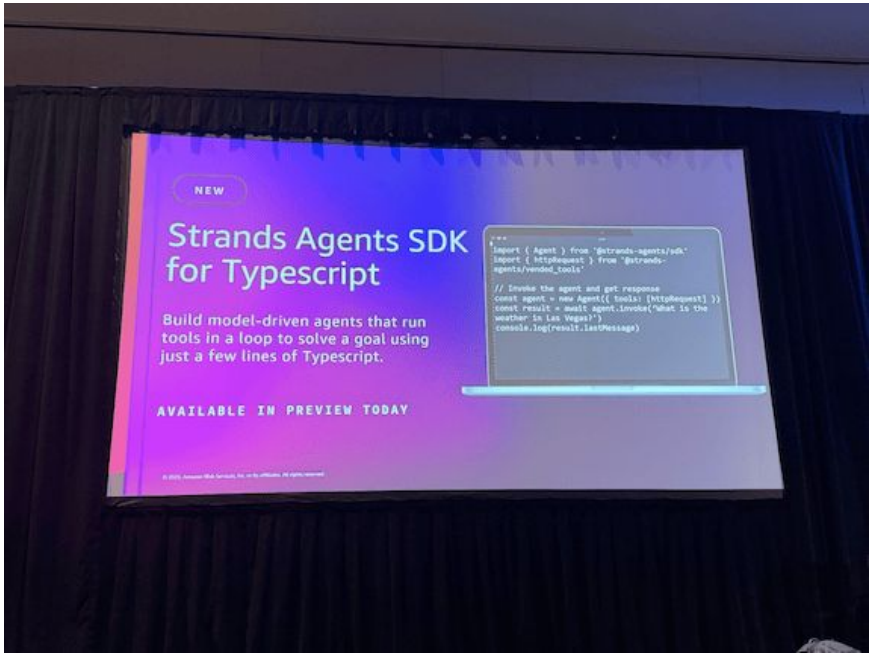
Capturing Streamed Data & Events

Async Iterators

Next Steps



https://dev.classmethod.jp/articles/reinvent2025-strands-agents-sdk-typescript-sop/



https://dev.classmethod.jp/articles/reinvent2025-strands-agents-sdk-typescript-sop/

Strands SOPs

- Standard Operating Procedures

```
# Code Assist

## Overview
This SOP guides the implementation of code tasks using test-driven development
principles, following a structured Explore, Plan, Code, Commit workflow.

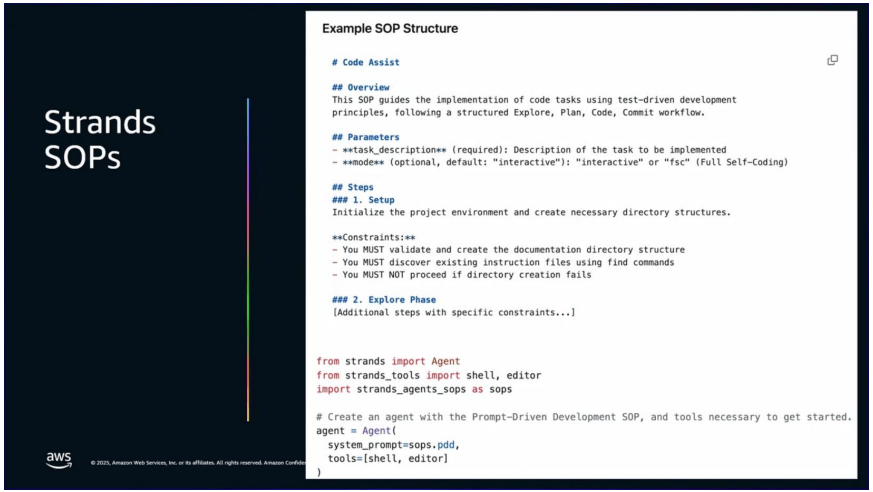
## Parameters
- **task_description** (required): Description of the task to be implemented
- **mode** (optional, default: "interactive"): "interactive" or "fsc" (Full Self-Coding)

## Steps
### 1. Setup
Initialize the project environment and create necessary directory structures.

**Constraints:**
- You MUST validate and create the documentation directory structure
- You MUST discover existing instruction files using find commands
- You MUST NOT proceed if directory creation fails

### 2. Explore Phase
[Additional steps with specific constraints...]
```

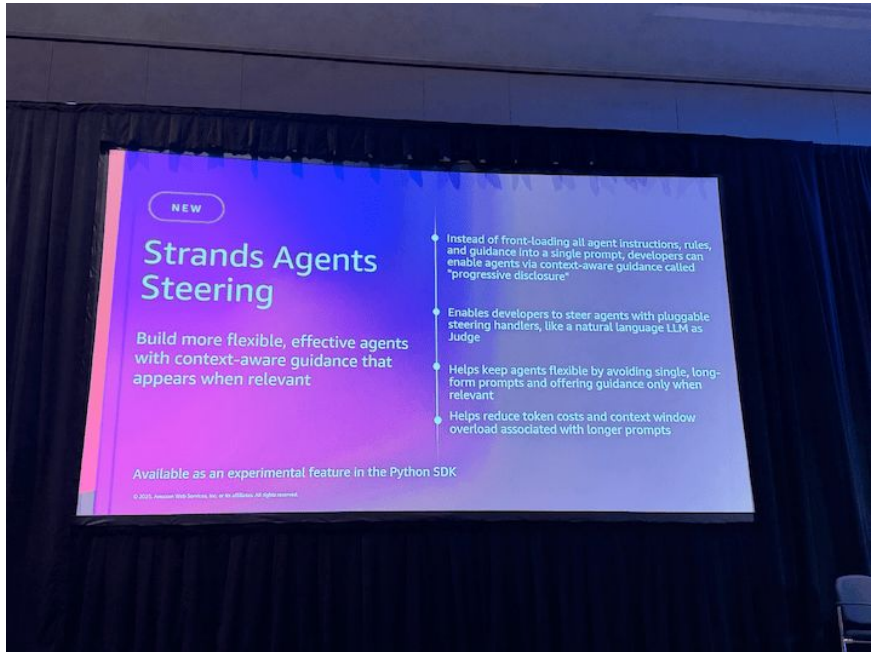
https://github.com/strands-agents/agent-sop



https://djn5cn31a5nkh.cloudfront.net/NzZOm-kaO94/251207_NzZOm-kaO94_korean_report.html

Strands Agents Steering

- Standard Operating Procedures



https://dev.classmethod.jp/articles/reinvent2025-strands-agents-sdk-typescript-sop/

Build, fine-tune & deploy AI models with SageMaker HyperPod CLI & SDK

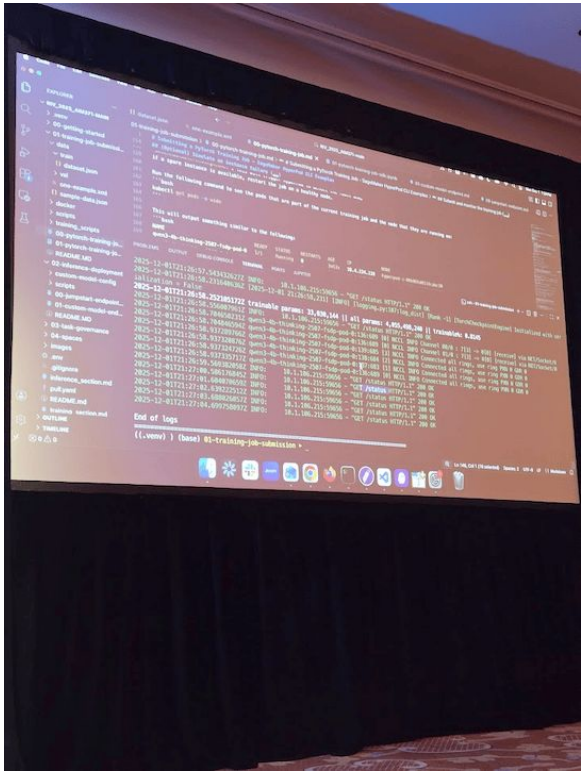
SageMaker HyperPod CLI

- EKS에 띄우는 SageMaker HyperPod 클러스터 관리 도구

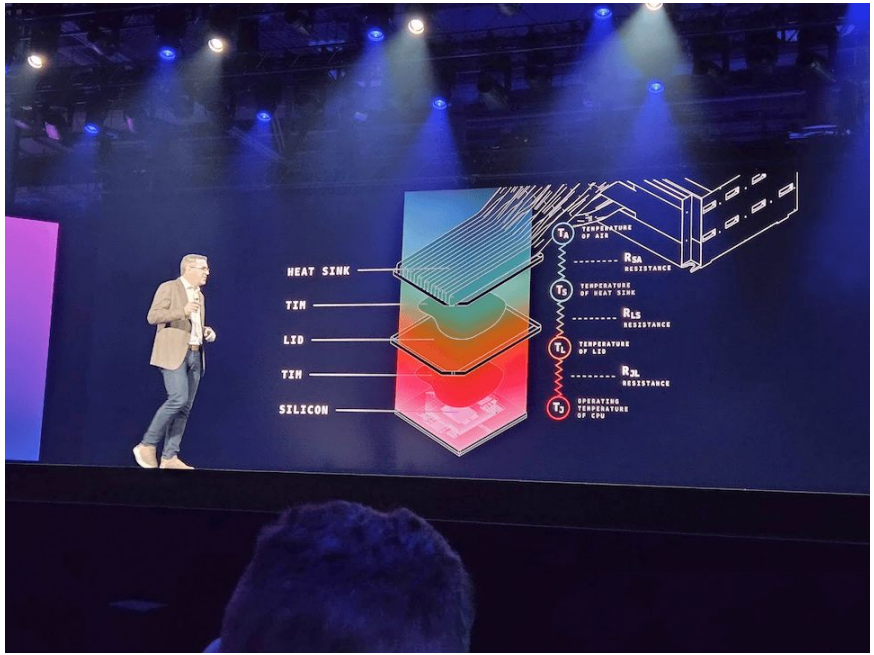

ATH321
Build, fine-tune & deploy AI models with SageMaker HyperPod CLI & SDK
Giuseppe A. Porcelli
he/him
Principal ML Solutions Architect
Amazon Web Services
Arun Nagarajan
he/him
Principal Software Engineer
Amazon Web Services
aws re:Invent

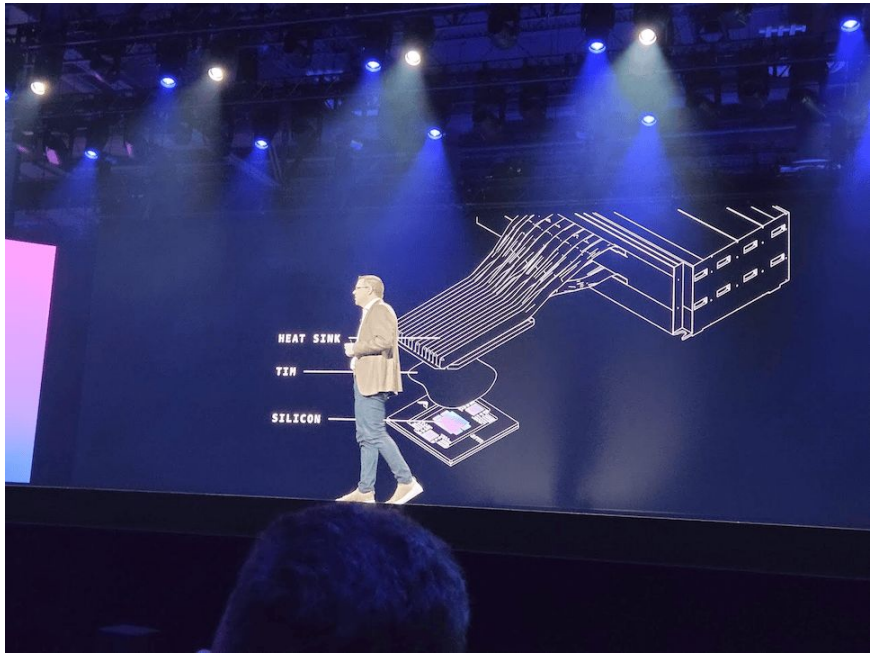

Amazon SageMaker HyperPod
Scale and accelerate generative AI model development across thousands of AI accelerators
Improved efficiency
Tools for maximizing compute resources utilization, advanced observability, and seamless cluster customization
Reduced time-to-train
Resilience features and distributed training libraries help reduce time to train by up to 40%
Lower costs
Less time invested in hardware maintenance and more efficient cluster engagement reduces FM training TCO
aws
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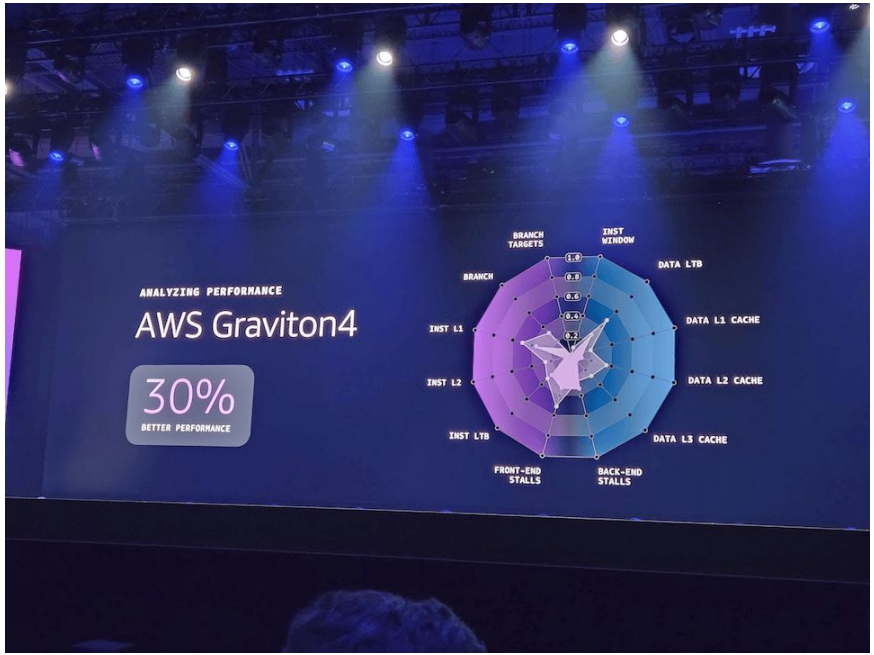

HyperPod benefits
Resilient
Proactively screen health of inbound nodes
Continuous cluster hardware monitoring
Automated repair and job resumption
Spare capacity dedicated to self-healing
Scalable
Single-spine node topology
Pre-configured EFA for optimal inter-node communication speeds
Flexible paths to securing compute capacity
Rapid cluster scale-up without performance degradation
Versatile
Broad GenAI software stack compatibility—with extensive collection of examples
Lifecycle scripts or Helm for streamlined cluster customization
Support for a variety of top accelerator instances such as Trn2, P6-B200, P5en, P5, P4de, etc.
Efficient
Task governance maximizes cluster utilization
Integrated SageMaker AI tools and libraries
Optimized recipes for starting FM training in minutes
HyperPod-managed tiered checkpointing
Advanced observability with unified visibility across AI model development tasks and compute resources
aws
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Learnings from developers adopting AWS Graviton at scale


HEAT SINK
TIM
LID
TIM
SILICON
TA
TEMPERATURE OF AIR
RSA
RESISTANCE
TS
TEMPERATURE OF HEAT SINK
RLS
RESISTANCE
TL
TEMPERATURE OF LID
RJL
RESISTANCE
TJ
OPERATING TEMPERATURE OF CPU


HEAT SINK
TIM
SILICON


ANALYZING PERFORMANCE
AWS Graviton4
30%
BETTER PERFORMANCE
BRANCH TARGETS
INST WINDOW
DATA LTB
DATA L1 CACHE
DATA L2 CACHE
DATA L3 CACHE
BACK-END STALLS
FRONT-END STALLS
INST LTB
INST L2
INST L1
BRANCH
1.0
0.8
0.6
0.4
0.2


NEW
AWS Graviton5
Our most efficient CPU ever
AVAILABLE IN PREVIEW TODAY
PCIe 6
DDR5

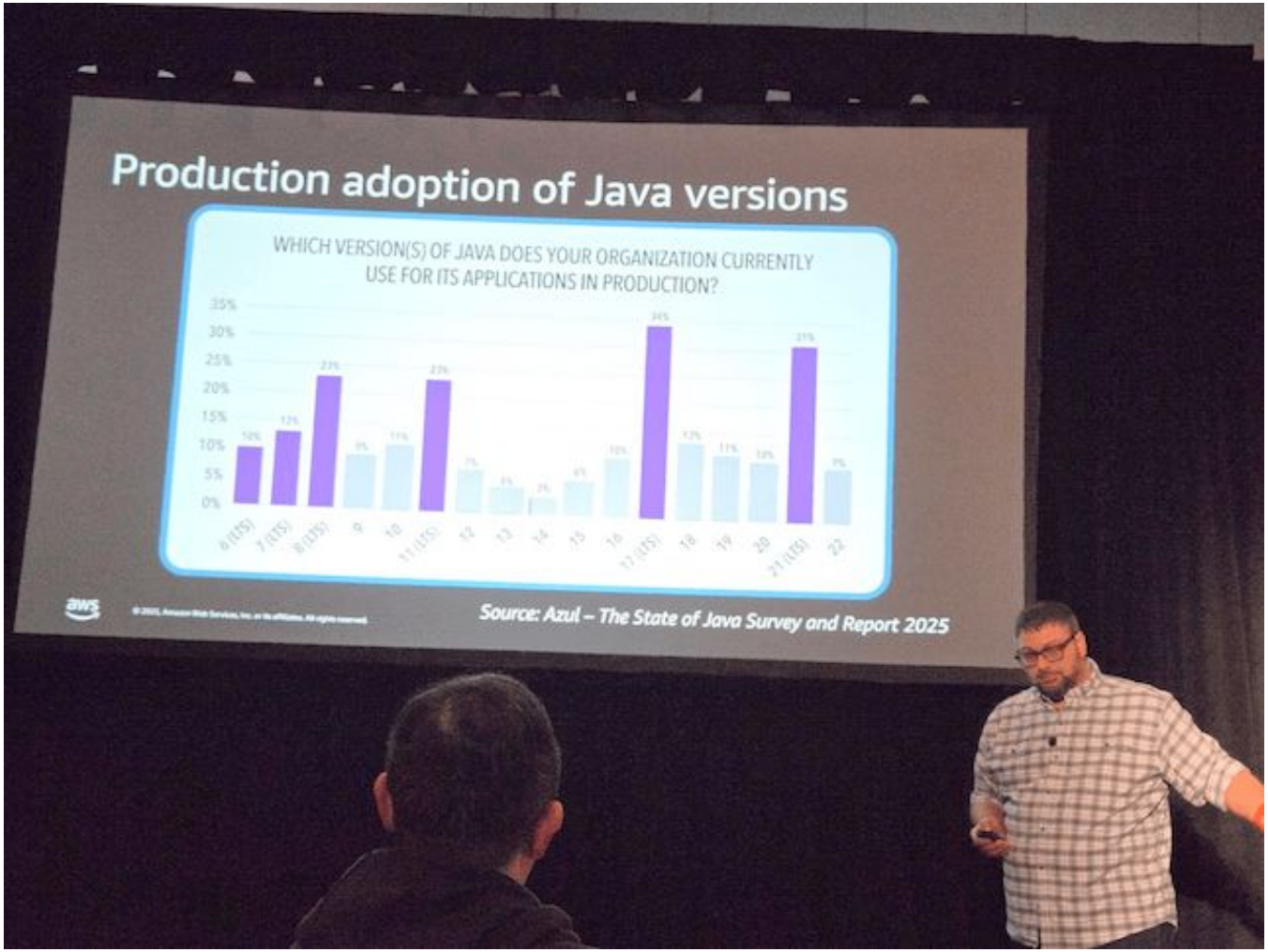

CMP223
Learnings from developers adopting AWS Graviton at scale
Jeff Underhill
Paul Reilly
Amazon EC2 Compute Specialist
Amazon EC2 Compute Specialist
re:Invent

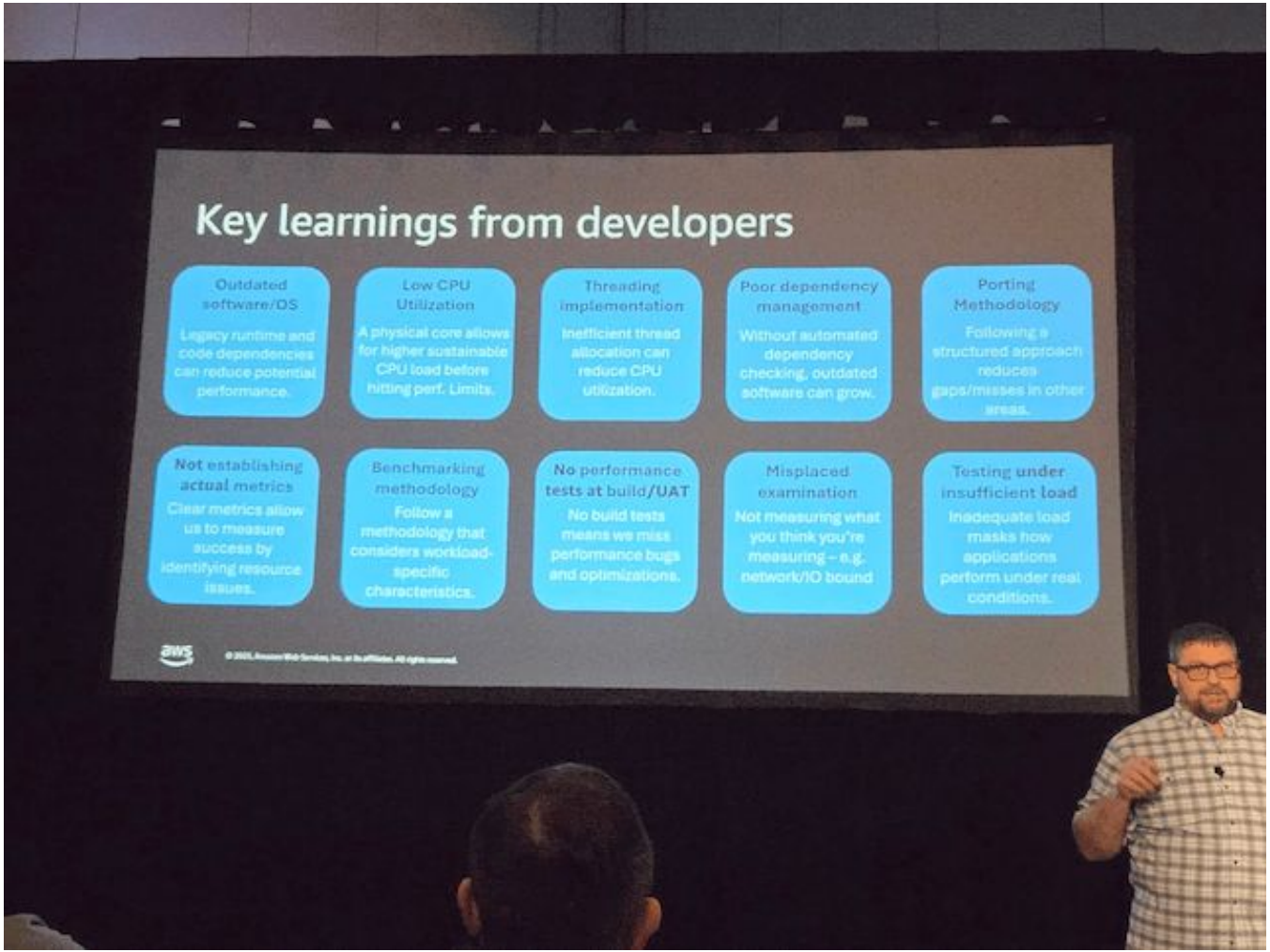

Graviton at scale
Paul Reilly
re:Invent

A brief history of Graviton

2018	2019	2022	2023
AWS Graviton	AWS Graviton2	AWS Graviton3	AWS Graviton4
			
Up to 45% cost reduction over comparable instances for scale-out workloads	Up to 40% better price performance than comparable instances	Up to 25% better compute performance and up to two times the floating-point performance of Graviton2	Up to 30% better compute performance, 50% more cores, and 75% more memory bandwidth than Graviton3

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Production adoption of Java versions
WHICH VERSION(S) OF JAVA DOES YOUR ORGANIZATION CURRENTLY USE FOR ITS APPLICATIONS IN PRODUCTION?
35%
30%
25%
20%
15%
10%
5%
0%
6 (EOS)
7 (EOS)
8 (LTS)
9
10
11 (LTS)
12
13
14
15
16
17 (LTS)
18
19
20
21 (LTS)
22
aws
Source: Azul – The State of Java Survey and Report 2025


Key learnings from developers
Outdated software/OS
Legacy runtime and code dependencies can reduce potential performance.
Low CPU Utilization
A physical core allows for higher sustainable CPU load before hitting perf. Limits.
Threading implementation
Inefficient thread allocation can reduce CPU utilization.
Poor dependency management
Without automated dependency checking, outdated software can grow.
Porting Methodology
Following a structured approach reduces gaps/misses in other areas.
Not establishing actual metrics
Clear metrics allow us to measure success by identifying resource issues.
Benchmarking methodology
Follow a methodology that considers workload-specific characteristics.
No performance tests at build/UAT
No build tests means we miss performance bugs and optimizations.
Misplaced examination
Not measuring what you think you're measuring – e.g. network/IO bound
Testing under insufficient load
Inadequate load masks how applications perform under real conditions.
aws


aws
EC2 Instance
Compute
Memory
Network
Storage
Generation
Series
Options
c8gn.2xlarge
Instance family
Instance size
Instance type
esc

EOD